

# THE SOLAR SYSTEM

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# 1 Our Planetary Neighborhood

## 1.1 The Sun and Its Planets

The Sun and its gravity-bound bodies form the Solar System [1]. See [NASA Solar System Exploration](#). In increasing distance, the planets are

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.

The planets form two groups:

### (I) Inner planets

- † Mercury, Venus, Earth, and Mars belong to this group.
- † They are comparatively small and have solid surfaces.
- † They are also called the terrestrial planets.

### (II) Outer planets

- † Jupiter and Saturn are gas giants.
- † Uranus and Neptune are commonly called ice giants.
- † All four outer planets possess ring systems.

The inner planets are small, dense, and relatively close to the Sun, whereas the outer planets are much larger and travel along considerably wider orbits.

## 1.2 Portraits of the Solar System

Figure 1 shows the Sun and eight planets.<sup>1</sup> Compare Mars in Figure 1e with Jupiter in Figure 1f.

# 2 Kepler's Laws of Planetary Motion

Kepler's three laws describe planetary motion [2].

## 2.1 First Law: Elliptical Orbits

A planetary orbit is elliptical, with the Sun at one focus:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a \geq b > 0, \quad (1)$$

Here  $a$  and  $b$  are the semi-axes. For focal distance  $c$ ,

$$\begin{aligned} c^2 &= a^2 - b^2 \\ e &= \frac{c}{a} \end{aligned} \quad (2)$$

Here  $e$  is the eccentricity. In polar form,

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}. \quad (3)$$

Equation (3) gives the orbital distance.

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<sup>1</sup>Sizes and distances are schematic.

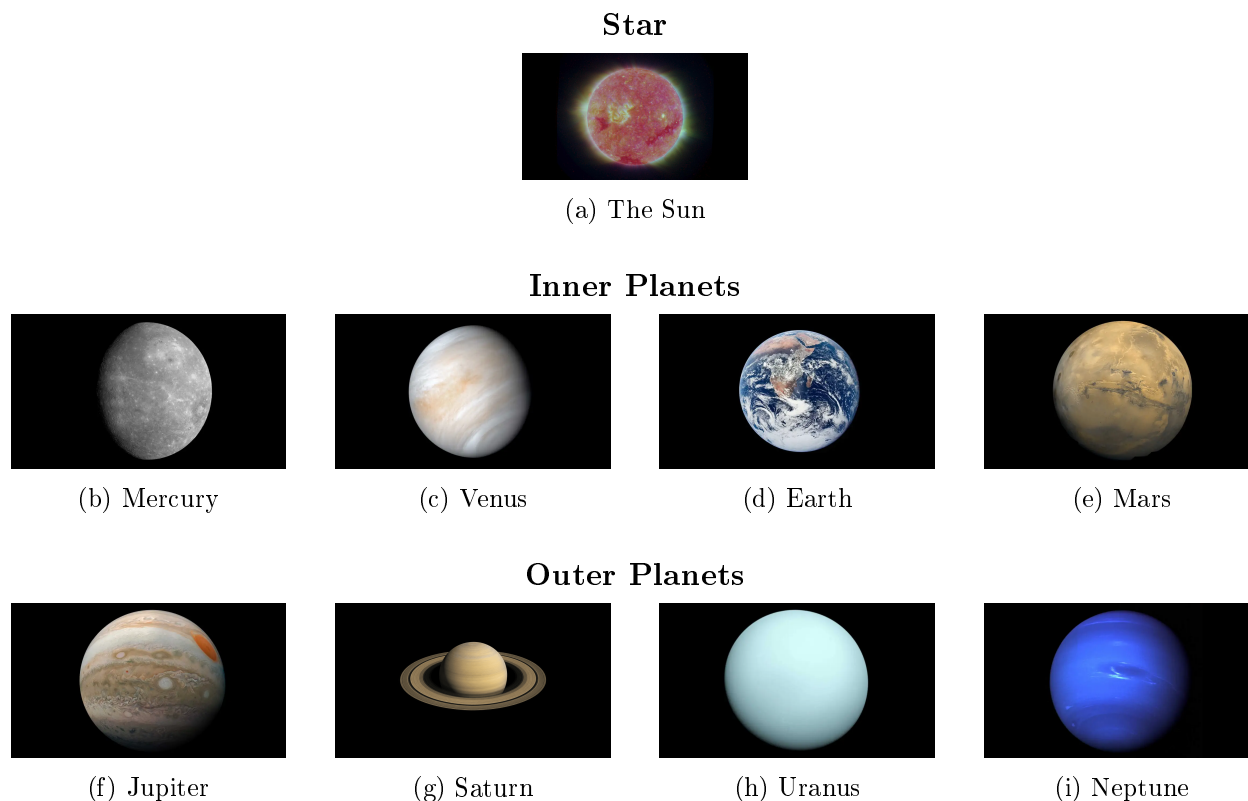


Figure 1: The Sun and the eight planets of the Solar System, arranged into the inner and outer planetary groups. The displayed sizes and distances are not to scale.

## 2.2 Second Law: Equal Areas in Equal Times

Equal times sweep equal areas:

$$\Delta t_1 = \Delta t_2 \implies \Delta A_1 = \Delta A_2. \quad (4)$$

Consequently, a planet moves

- ◊ faster when it is closer to the Sun, and
- ◊ slower when it is farther from the Sun.

## 2.3 Third Law: Orbital Period and Distance

For planets orbiting one star,

$$T^2 \propto a^3. \quad (5)$$

Newtonian form:

$$T^2 = \frac{4\pi^2}{GM_\odot} a^3,$$

Here  $M_\odot$  is the solar mass.

## 2.4 An Earth–Mars Comparison

Using Earth as reference,

$$a_E = 1 \text{ AU} \quad T_E = 1 \text{ year}$$

where  $1 \text{ AU} \approx 1.496 \times 10^{11} \text{ m}$ .<sup>2</sup> For Mars,  $a_M \approx 1.524 \text{ AU}$ , so

$$\begin{aligned} \frac{T_M^2}{T_E^2} &= \frac{a_M^3}{a_E^3}, \\ T_M &= T_E \left( \frac{a_M}{a_E} \right)^{3/2} \\ &\approx (1.524)^{3/2} \approx 1.88 \text{ years.} \end{aligned}$$

Thus  $T_M \approx 1.88 \text{ years}$ .

## 3 A Relativistic Extension

General relativity extends Kepler’s model.<sup>3</sup> Einstein’s equations are

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (6)$$

For spherical mass  $M$ ,

$$\begin{aligned} ds^2 &= - \left( 1 - \frac{r_s}{r} \right) c^2 dt^2 + \left( 1 - \frac{r_s}{r} \right)^{-1} dr^2 \\ &\quad + r^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad r_s = \frac{2GM}{c^2}. \end{aligned} \quad (7)$$

Free fall satisfies

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (8)$$

Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}. \quad (9)$$

**Proposition 3.1** (Weak-field circular orbit). *For a circular orbit of radius  $r$ ,*

$$\left( \frac{v}{c} \right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}. \quad (10)$$

*Proof.* The circular-orbit balance  $v^2/r = GM/r^2$  gives  $v^2 = GM/r$ . Divide by  $c^2$  and use  $r_s = 2GM/c^2$ .  $\square$

See Proposition 3.1.

## 4 The Outer Planets

The outer planets are Jupiter, Saturn, Uranus, and Neptune [3]. The first two are *gas giants*; the last two are *ice giants*. Table 1 uses scientific and relativistic notation.<sup>4</sup>

Table 1 compares the four giant planets.

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<sup>2</sup>AU denotes the mean Earth–Sun distance.

<sup>3</sup>Planetary corrections are small but measurable.

<sup>4</sup>Values are rounded for typesetting practice.

Table 1: Orbital and relativistic parameters of the outer planets

| <b>Planet</b> | <b>Orbital parameters</b> |                       | <b>Relativistic notation</b>           |                              |
|---------------|---------------------------|-----------------------|----------------------------------------|------------------------------|
|               | $a$ (AU)                  | $e$                   | $\varepsilon = \frac{1}{ac^2}GM_\odot$ | $\beta = \sqrt{\varepsilon}$ |
| Jupiter       | 5.204                     | $4.89 \times 10^{-2}$ | $1.90 \times 10^{-9}$                  | $4.36 \times 10^{-5}$        |
| Saturn        | 9.583                     | $5.65 \times 10^{-2}$ | $1.03 \times 10^{-9}$                  | $3.21 \times 10^{-5}$        |
| Uranus        | 19.191                    | $4.72 \times 10^{-2}$ | $5.14 \times 10^{-10}$                 | $2.27 \times 10^{-5}$        |
| Neptune       | 30.070                    | $8.6 \times 10^{-3}$  | $3.28 \times 10^{-10}$                 | $1.81 \times 10^{-5}$        |

## References

- [1] J. Kepler, *Astronomia Nova*. Heidelberg, 1609.
- [2] National Aeronautics and Space Administration, “Orbits and kepler’s laws,” 2024.
- [3] National Aeronautics and Space Administration, “About the planets,” 2026. [Solar System Exploration](#).